

Outcomes of Property Tax Regressivity

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Abstract

Tax assessments are regressive with respect to home sale price for owner-occupied single-family homes in the United States: assessors value cheaper homes high and expensive homes low relative to what they actually sell for. I ask whether that regressivity translates into adverse outcomes for owners of low-value homes when a mandatory assessment raises their tax bills. North Carolina requires countywide reappraisal at least every eight years, so Mecklenburg County's assessed values are frozen between reappraisals and jump at each one, producing a tax shock whose timing is statutory rather than responsive to any home's value. I assemble a parcel-year panel of administrative tax liabilities and transactions for the county's owner-occupied stock, sorting homes into value quartiles by their arm's-length sale prices in constant dollars, fixed before the window opens. The comparison is a triple difference: the cheapest quartile against the rest, and Mecklenburg against a weighted combination of thirty-three annually assessing counties, before the assessment against after. Counties qualify on conditions computed from their own records, and the same conditions govern the placebo distribution. Foreclosure filings among the cheapest homes rise by 0.20 percentage points and their sale prices are unchanged, though neither effect is unusual against that distribution. A second design instruments the change in a parcel's tax bill with an indicator for the reappraisal year, interacted with quartiles. Because reappraisal corrects assessments most where they had drifted furthest, the shock is graded by value: the cheapest quartile's bill rises by 25.6 percentage points against the most expensive quartile's 3.5, and foreclosure filings rise by 0.75 percentage points among the cheapest homes and by 0.13 among the most expensive, both significantly. Regressive assessment concentrates the burden through the size of the correction rather than through any greater sensitivity to it, and the distress that follows falls on the owners least able to absorb it.

1 Introduction

Tax assessments are regressive with respect to home sales price for owner-occupied single-family homes in the United States (McMillen & Singh 2019). This stylized fact has been linked to differences in assessment appeal

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rates and transaction frequency between high- and low-value homes (Amornsiripanitch, 2020; Avenancio-León and Howard, 2022), and translates to higher effective tax rate-for owners of cheaper housing. This paper asks whether this disparity in tax burdens is associated with sales price discounts and increased foreclosure filings for owners of cheaper homes.

One empirical challenge in studying the effects of property taxation on homeowner outcomes is that tax amounts change by only a small amount from year to year, on average. Without a large change in the tax bill in any given year, adverse outcomes for homeowners are not credibly attributable to property tax burdens. I address this by choosing a setting with infrequent assessments, Mecklenburg County, North Carolina. Mecklenburg homeowners face large jumps in their tax assessments every eight years and frozen assessed values in between. The bill is the product of the assessed value and the statutory rate, so it still drifts between reappraisals as the rate is reset each year, but those movements are small and fall on every home alike; the reappraisal is what delivers a large change in what an owner owes, and one that differs sharply across the value distribution.

A second obstacle to a causal estimate is that property tax bills and home sales prices are jointly determined. Assessed values are correlated with sales prices, so appreciation raises tax bills through assessment. A regression of a sales price discount on the tax bill therefore conflates the effect of taxes on values with the reverse. The same setting breaks this simultaneity through two features of the North Carolina property tax. assessments are countywide and statutorily timed, so the discrete tax shock they generate does not respond to any individual home's value.

Still, owners of cheaper housing may be more liquidity-constrained and therefore be more adversely affected by any changes in the cost of living relative to owners of expensive housing. If these changes occur along the same timeline as assessments, changes in outcomes for owners of cheaper housing relative to those of expensive housing could be confounded by these baseline differences in liquidity. To further isolate the effects of property tax bills from those of other confounders, I use a triple-differences approach comparing outcomes for Mecklenburg homeowners to those of homeowners in a synthetic county built from annually assessing counties. I compare differences in outcomes for the owners of bottom-quartile homes in the two counties before and after a Mecklenburg assessment year. The resulting effects can then be more credibly attributed to the large changes in tax bills experienced by owners of cheap housing after an assessment year.

I use a second empirical design that directly measures the impact of changes in tax bills on homeowner outcomes. Because these tax shocks and sale prices are jointly determined, I instrument for the change in the tax bill with an indicator for the assessment year. Because the timing of assessments is predetermined based on longstanding state mandates, it can move sales prices only through changes in tax bills. Using this approach allows for a percentage point comparison between changes in tax bills and changes in sales prices

or in the probability of a transaction or foreclosure filing.

My study uses parcel-level assessor and transaction data from ZTRAX for all counties of interest, allowing me to capture the assessment and sales values of each home. Mecklenburg County’s tax assessor office provides uniquely detailed data on historical tax bills for all parcels from 2012 to 2022, allowing me to capture bill changes around its 2019 assessment. I first document regressivity in property taxation across Mecklenburg County’s home value distribution, finding that percent increases for the lowest value quartile are six times the size of the increases in the highest quartile in the assessment year, on average.

In the triple difference against that synthetic county, bottom-quartile homes sell at roughly an 8 percent discount and see 0.3 percentage points more foreclosure filings in the years following an assessment, though neither effect is unusual against the distribution of placebo effects across donor counties. Results are smaller in magnitude but qualitatively similar when using the synthetic control group. In a second design instrumenting for the tax bill change with a assessment year indicator interacted with pre-period sales price quartiles, I find that bottom quartile homes sold at a 15 percent discount post assessment. This sales price discount monotonically shrinks in home value quartile and reverses for the highest quartile, who sees a 15 percent sales price premium post assessment.

This paper uses an unusually detailed dataset of historical tax bills and transaction prices for Mecklenburg homes, and a unique quasi-experimental design to understand the heterogeneous effects of property taxation. A rich literature documents assessment regressivity (Amornsiripanitch, 2024; Avenancio-León and Howard, 2022) and a smaller one links tax bills to default and foreclosure, but none estimates whether the distress and price consequences fall differentially on low-value homes.¹ The two designs together locate the regressive burden in a specific margin: it surfaces as financial distress for the cheapest owners, consistent with thinner equity buffers absorbing the shock through household balance sheets.

2 Background

2.1 Property Tax Regressivity

A stylized fact of the assessment literature is that property tax bills are regressive with respect to a property’s sales price. This is because assessed values are weakly correlated with the sales prices. Assessed values are deliberately set below and rise less than one-for-one with sales prices in part because wealthier owners appeal more often and mass-appraisal models fit sparse, low-turnover neighborhoods poorly (Amornsiripanitch, 2024). Assessment rates—the ratio of assessed value to the sale price of a property—are in turn higher for

¹The closest work studies the expiration of a Philadelphia abatement and finds mortgage delinquency rose sharply, but speaks to new construction rather than to the distributional incidence of a countywide tax structure (Coste, 2024).

lower-priced homes.

Two other significant components of property tax bills are statutory tax rates and homestead exemptions. Statutory tax rates (commonly known as millage rates) are set at the jurisdictional level, smaller than the county level, while assessments are done by the county tax assessor office. The assessment literature has shown that regressivity is present at both the county and jurisdictional level (Amornsiripanitch, 2024; Avenancio-León and Howard, 2022). Homestead exemptions are lump sum deductions one can apply to their assessment value to reduce their tax base. Authors McMillen and Singh (2020) show that while homestead exemptions reduce tax liabilities for lower value homes more, they also do not change the regressivity artifact.

2.2 Mecklenburg County, North Carolina

The setting for my study is a fast-growing county with infrequent assessment cycles Mecklenburg County, North Carolina. assessment schedules are bounded by state law but set locally in most states, including North Carolina, where the statute requires a countywide reappraisal at least every eight years.²³ Mecklenburg County contains Charlotte and six other jurisdictions, all subject to annually changing millage rates. In this way, millage rates can be increased during nonassessment years to increase their revenue despite frozen assessment values.⁴

This setting creates useful variation in the tax burden. Because assessed values are frozen between reappraisals while sale prices continue to move, a home’s assessment falls further behind what it would fetch on the market with every year that passes, and the whole gap closes at once when the county reappraises. That gap is wider in proportional terms for cheaper homes, so the reset at each reappraisal falls hardest on them. Figure 1 shows the pattern in the cross-section, comparing the assessor’s value against the price for homes assessed and sold in the same year.

3 Empirical Approaches

I ask how the regressive nature of property taxation affects homeowners across the home value distribution. North Carolina mandates countywide reappraisal at least every eight years (NCGS § 105-286); Mecklenburg assessed in 2011, and 2019. My two research designs exploit variation in tax bills induced through years since

²³This was recently changed to every four years after the 2019 assessment for Mecklenburg, County but the state mandate remains the same as of 2026

³Montana, Nebraska, North Carolina, North Dakota, Ohio, Oregon, Rhode Island, and Tennessee also have assessment interval extending into the six-to-ten-year range.

⁴While this subcounty dynamic millage rates provides another avenue through which property tax regressivity can enter at the county level if there is sorting of cheaper homes into jurisdictions that raise millage rates more aggressively, this is shown not to be the case in Table A2

last assessment. Year-over-year tax bills changes are moderate in non-assessment years and rise sharply at assessments, for the average homeowner. This produces quasi-experimental variation in discrete tax shock at each assessment. Across the home value distribution, bottom-quartile homeowners face the largest changes in their tax bills at assessment while top-quartile homeowners see much smaller changes.

3.1 Triple-Differences

A difference in differences approach in this setting would compare outcomes for bottom-quartile homes—treated with the highest dosage of the tax shock—to those of all other homeowners. I include an additional comparison to control for potential differences between housing quartiles that might confound the effects of the tax shock. If housing market or employment cycles affect quartiles differently and their timing is correlated with Mecklenburg’s assessment schedule, the effect may be confounded by omitted variable bias. To mitigate this, I employ a third comparison: counties with housing markets similar to Mecklenburg’s that assess annually, so that their tax bills move gradually while their similarities help control for between-quartile differences in housing outcomes.

The comparison unit is built from counties that assess annually, weighted so that it reproduces Mecklenburg’s pre-assessment path; Section 3.2 sets out how, and why a single county will not serve. Where assessment is annual, assessed value tracks market value continuously, so no appreciation accumulates unrecognized between reappraisals and no discrete correction is ever owed. Owners in such a county absorb the same underlying house-price movements as Mecklenburg owners, but they meet them as a sequence of small annual adjustments rather than as one step at a assessment date. An annually assessing county therefore experiences nothing at the Mecklenburg event dates, while remaining exposed to the regional housing and labor cycles that would otherwise confound the comparison. The residual within-quartile differences in housing outcomes can then be more credibly attributed to tax changes.

The first design asks whether the effects of an assessment concentrate on the lowest-value homes, the group whose assessments are corrected most sharply and for which a tax change passes through most fully. I estimate a stacked static triple difference that compares the cheapest homes to the rest, and the assessing county to the annually assessed control, over the window around each assessment:

$$Y_{i,e,t} = \beta (\text{Post}_t \times B_i \times C_i) + (\text{two-way interactions and main effects}) + \mathbf{X}_i' \boldsymbol{\delta} + \alpha_{c(i),e} + \varepsilon_{i,e,t}, \quad (1)$$

where $Y_{i,e,t}$ is the log sale price, the log price per square foot, a transaction indicator or a foreclosure-filing indicator; the price outcomes are defined on parcel-years carrying a sale and the two rates on every parcel-year in the window. Here $e \in \{2011, 2019\}$ indexes the assessments over which the sample is stacked,

$\text{Post}_t = \mathbf{1}[\tau_{ie} \geq 0]$ for event time $\tau_{ie} = t - e$, C_i marks the assessing county and B_i the bottom quartile of the within-county value distribution, cut on constant-dollar sale prices fixed before the window opens. The vector $\mathbf{X}_i$ collects structure characteristics, $\alpha_{c,e}$ are comparison-unit by event fixed effects, and standard errors are clustered by parcel. Each event’s window opens at $\tau = -4$ and closes when the next reappraisal lands.

The coefficient β is the object of interest: how the cheapest treated homes move after an assessment, net both of what happened to the upper quartiles of the same county and of what happened to the cheapest homes of the comparison unit. The triple difference removes level differences between value groups common to both units and shocks common to both value groups within a unit. Identification rests on parallel trends in the remaining contrast: absent the assessment, the gap between the cheapest homes and the rest would have moved in Mecklenburg as it did in the comparison unit.

That assumption is the whole design, and it is why the comparison unit is constructed rather than chosen. Section 3.2 builds it as a weighted combination of annually assessing counties fitted to Mecklenburg’s pre-assessment path, and the effect it reports in (4) is the sample analogue of β in (1).

Figure ?? shows the two rate outcomes by value quartile in Mecklenburg, with the two assessment years marked. Both series are ordered by value throughout, the cheapest quartile sitting above the rest in every year, which is the ordering the design exploits. What the figure cannot supply is a counterfactual for how that ordering would have moved absent an assessment, and no single county provides one: the comparison unit is constructed in Section 3.2 instead.

3.2 Synthetic control

A single control county carries the whole weight of the parallel-trends assumption, and no county in the region satisfies it: the annually assessing counties nearest Mecklenburg in size moved through the housing crisis on visibly different schedules from it, so their cheapest homes were not a counterfactual for Mecklenburg’s. The remedy is not a better single county but a weighted combination of many, chosen so that the combination reproduces Mecklenburg’s pre-assessment behaviour more closely than any one county does.

Every county with identifiable owner-occupants and a complete transactions enters the candidate set. Which of them can serve as a control is then decided by five conditions. First, the county must assess annually, so it absorbs no discrete correction of its own inside a Mecklenburg window. Second, its deed recordings must credibly track its housing market rather than fill in over time.⁵ Third, enough of the county’s stock must be sorted into value quartiles by prior sales. Fourth, each quartile-year cell inside the

⁵Coverage in several states improves across the sample for administrative reasons, which produces a rising sales series that has nothing to do with housing. Both parts are required: counties whose sales fell during the crisis show low volume too, and they are kept.

window must hold enough sales to compute an outcome. Fifth, owner-occupants must be identifiable, since a universe containing rentals and second homes is not the universe the design compares.

The screen governs the placebo distribution as well as the donor pool. Inference compares the treated county’s post-assessment divergence to the divergence of each donor treated as if it had been assessed. A county whose deed record is filling in shows a compressed pre-assessment path and a rising post-assessment one, so it registers a large apparent effect for reasons unrelated to any assessment and displaces Mecklenburg in the ranking. Excluding such a county from the weights while retaining it as a placebo would leave the comparison distorted.

The weights are fitted to Mecklenburg’s pre-assessment trajectory rather than to its level. This matters because the estimator constrains weights to be non-negative and to sum to one, so the synthetic county is confined to the range the donors span. Imposed on levels, this penalizes a donor whose gap between the cheapest quartile and the rest moves exactly as Mecklenburg’s does but sits below it. Since the triple difference identifies from how that gap moves and not from where it sits, matching on levels selects donors on the feature the design removes.

Let Y_{jqt} denote the outcome in county j , value quartile q , and year t , with $j = 0$ the assessing county and $j = 1, \dots, J$ the donors, and let $\mathcal{T}^-$ collect the pre-assessment years. Write each series as a deviation from its own pre-assessment mean,

$$\tilde{Y}_{jqt} = Y_{jqt} - \frac{1}{|\mathcal{T}^-|} \sum_{s \in \mathcal{T}^-} Y_{jqs}, \quad (2)$$

so that only the shape of each county’s path survives. Weights are then chosen to reproduce Mecklenburg’s four quartile paths at once,

$$\mathbf{w}^* = \arg \min_{\mathbf{w} \in \Delta^J} \sum_{q=1}^4 \frac{1}{\sigma_q^2} \sum_{t \in \mathcal{T}^-} \left(\tilde{Y}_{0qt} - \sum_{j=1}^J w_j \tilde{Y}_{jqt} \right)^2, \quad \Delta^J = \left\{ \mathbf{w} : w_j \geq 0, \sum_j w_j = 1 \right\}, \quad (3)$$

where σ_q is the standard deviation of the deviations in quartile q across counties and pre-assessment years, so that a quartile does not dominate the objective by moving on a larger scale. Subtracting each county’s own mean in (2) is what restores the correspondence between what the estimator matches and what the design uses (Ferman et al., 2020). Stacking the four quartiles in (3) means a donor earns its weight by resembling Mecklenburg across the whole value distribution rather than by matching a single difference of averages, which two counties with quite different quartile paths can do by cancellation. That the weights are fitted on four series while the effect below is read on one is not an inconsistency: the fitted series are this design’s predictors, and the canonical estimator likewise chooses weights on lagged outcomes and covariates together

while reporting a single outcome (Abadie et al., 2010).

Equation (3) is solved separately for every outcome and every assessment, so each cell of the results has its own donor pool. Nothing forces the counties that reproduce Mecklenburg’s price path in 2011 to be the counties that reproduce its foreclosure path in 2019, and in practice they are not: the composition of the synthetic county turns over substantially between cells. This is the original form of the method (Abadie et al., 2010) and it is the appropriate one here, since a single weight vector imposed across outcomes would fit each of them worse. Two consequences follow for how the results are read. Each estimate is reported alongside the donors that produced it, because a coefficient resting mostly on one county is a different claim from one resting on many. And where two cells point the same way, they do so from separately chosen comparison groups, so the agreement is corroboration rather than a shared construction reproducing itself.

The effect is read on the same gap the triple difference uses. With $G_{jt} = Y_{j1t} - \frac{1}{3} \sum_{q=2}^4 Y_{jq t}$ and $\tilde{G}_{jt}$ its deviation from the pre-assessment mean,

$$\hat{\tau} = \frac{1}{|\mathcal{T}^+|} \sum_{t \in \mathcal{T}^+} \left(\tilde{G}_{0t} - \sum_{j=1}^J w_j^* \tilde{G}_{jt} \right), \quad (4)$$

the average post-assessment divergence between the treated county’s gap and its synthetic counterpart, net of any divergence already present before the assessment.

Feasibility of estimation is determined by asking whether Mecklenburg’s trajectory lies within the range the donors describe in every pre-assessment year and whether one fixed set of weights can track it. Cells that fail are reported as infeasible rather than estimated, because a synthetic county that cannot follow Mecklenburg before the assessment will appear to diverge from her afterwards for reasons that have nothing to do with the assessment.

3.3 Instrumental-Variables Estimates by Home Value

The triple difference identifies the effect of a reappraisal. It does not say how much of that effect travels through the tax bill, and it does not weigh the response of each group against the size of the bill increase that group actually received. The second design puts the bill itself on the right-hand side. It uses Mecklenburg alone, because the administrative levy is recorded there and would have to be imputed for the control county, and an imputed regressor has no place in a first stage. The cross-county comparison stays where it belongs, in Section 3.1.

Reappraisal resets every assessed value in the county at once, and it does not reset them proportionally. Assessors value cheaper homes generously against what those homes sell for and expensive homes conservatively, so a reappraisal that corrects accumulated drift corrects most where the drift ran furthest. Bills are

not otherwise still. They drift with the millage rate every year, in both directions: the county-wide change was -2.3 percent in 2012, $+2.0$ percent in 2013, and -4.1 percent in 2022. What distinguishes a reappraisal year is not that the bill moves but that it moves very differently across the value distribution. In 2019 the cheapest quartile’s bill rose 24.1 percent and the most expensive quartile’s 3.7, a spread of 20.4 percentage points; in no other year of the panel does that spread reach 2 points, and in most it is a fraction of one. A millage change is set at the jurisdiction level and falls on every quartile alike, so it shifts all four bills together and is absorbed by the fixed effects; reappraisal is what separates them. That separation is what this design uses.

Let ΔT_{it} be the percent change in parcel i ’s bill between $t-1$ and t , and let Z_t indicate a reappraisal year. Because the endogenous variable is a flow whose quartile gradient opens up in one year and is negligible in the rest, the instrument is the indicator for that year rather than a count of years since the last reappraisal: elapsed time predicts the level of the assessment, not the differential jump in the bill. Writing $Q_{iq} = \mathbf{1}[i \text{ in quartile } q]$ for the four quartiles of the within-county value distribution, cut on the same constant-dollar sale prices used in Section 3.1,

$$\Delta T_{it} \cdot Q_{iq} = \pi_q (Z_t \cdot Q_{iq}) + \mathbf{X}_{it}' \boldsymbol{\delta}_q + \alpha_{q,c(t)} + \varepsilon_{it}, \quad (5)$$

$$Y_{it} = \sum_{q=1}^4 \theta_q (\widehat{\Delta T}_{it} \cdot Q_{iq}) + \mathbf{X}_{it}' \boldsymbol{\beta}_q + \alpha_{q,c(t)} + u_{it}, \quad (6)$$

where $\mathbf{X}_{it}$ collects the tract house-price index, the county unemployment rate, and structure characteristics, $\alpha_{q,c}$ is a quartile by reappraisal-cycle fixed effect, and standard errors are clustered by census tract.

Every control and every fixed effect is indexed by quartile. Partitioning the endogenous variables and the instruments alone would not separate the four blocks, because controls held common across quartiles let each group use information from the others; interacting the controls as well makes each quartile’s estimate exactly the estimate its own observations support, which is what the inference below requires. The model carries no calendar-year fixed effects since reappraisal is countywide, so the reappraisal indicator is a function of the year alone and would be absorbed, taking one of its four quartile interactions with it and leaving three instruments for four endogenous regressors. Cycle fixed effects, the house-price index, and the unemployment rate absorb the county-wide path instead. This is the design’s principal cost relative to the triple difference, which nets out common time shocks with a control county rather than with covariates, and the estimates below should be read with that substitution in view.

4 Data

I assemble parcel-level data on tax liabilities, transactions, and property characteristics for the universe of owner-occupied homes in the treated county and in each comparison county. The Mecklenburg panel runs from 2011 through 2022 and spans two assessments, in 2011 and 2019. The comparison counties are built to the same specification from the same transaction source, so that a parcel-year means the same thing in every county and outcomes are measured identically across them. What differs by county is how owner-occupancy can be identified, discussed below.

Data on property tax amounts comes from the Mecklenburg County’s Open Mapping Tax Parcel CAMA Data.⁶ The records are a parcel-by-year panel of the county assessor’s administrative records. In addition to the tax levy, the records include assessed land and building values, property characteristics, year built, acreage, and land use. They also provide geographic identifiers including owner name, mailing address, assessor neighborhood and tax district codes. I treat a parcel as owner-occupied when the owner’s mailing address on the assessor’s roll matches the property’s own address, the standard administrative test for a primary residence.

Owner-occupancy cannot be identified the same way everywhere, because the transaction source is stripped of owner names and mailing addresses and counties differ in what they publish. Where a state grants a homestead exemption only on a primary residence, as Florida does, the exemption recorded against the parcel is a statutory indicator of owner-occupancy and I use it directly. Where a county publishes the owner’s mailing address alongside the property address, I apply the Mecklenburg test to those two fields. Where neither is available, the county cannot enter the comparison group on the same footing as the rest and is set aside rather than admitted under a different definition. Each parcel carries a record of which basis was used, so the composition of every county’s sample is recoverable.

I combine these tax data to transaction records from the Zillow Transaction and Assessment Database (ZTRAX, ICPSR Study 39652) using the mailing address as the merge key. The transaction records includes all transactions since 1990 with detail on sales price, deed type, mortgage amounts. I express every dollar amount in 2025 dollars using the annual-average consumer price index for all urban consumers.⁷

The resulting dataset links the assessor’s tax panel to transaction detail in each parcel-year. I sort homes into value quartiles using the median of a parcel’s arm’s-length sales between 1998 and 2011, each expressed in constant dollars before the median is taken. Averaging over a parcel’s sales rather than reading its most recent one alone removes the idiosyncratic noise carried by any single transaction, which matters for the two homes in five that sold more than once over the period. The window closes before the first assessment

⁶<https://mecksusftp001.blob.core.windows.net/gis/OpenMapping/index.html>

⁷<https://fred.stlouisfed.org/series/CPIAUCSL>

under study, so a home's place in the value distribution is fixed before the sample window opens and cannot respond to the tax changes I examine. Extending it forward would enlarge the sample, but only by admitting sales whose prices the assessment itself moved, which would sort a home into a value group partly because of the treatment it received. The treatment variable is the annual change in the tax bill. Outcomes are the probability of transacting, the log sale price and log price per square foot on transacting parcel-years, and indicators for a foreclosure filing and for a completed distressed sale in the parcel-year. Table 1 reports means by value quartile. Homes in the cheapest quartile are smaller and older than the rest of the stock, but they turn over at a similar rate and face a similar millage rate, while their foreclosure filing rate is more than three times that of the most expensive quartile.

Table 1 describes the resulting estimation sample, quartile by quartile. The quartiles are cut within the county on constant-dollar sale prices, so they partition its own stock rather than a national distribution, and the cheapest quartile is the group on which the assessment falls hardest.

Table 1: Summary statistics, Mecklenburg County.

Variable	Mean by value quartile			
	Q1	Q2	Q3	Q4
<i>Panel A. Parcel-year panel, 2011–2022</i>				
Assessed value (2025\$)	183,011 (129,519)	214,881 (102,147)	309,626 (132,587)	600,277 (341,173)
Millage rate (per \$100)	1.143 (0.159)	1.139 (0.159)	1.122 (0.159)	1.123 (0.156)
Annual property tax (2025\$)	2,031 (1,341)	2,400 (1,065)	3,419 (1,413)	6,683 (3,853)
Change in tax bill (%)	2.251 (10.209)	1.790 (8.223)	1.163 (6.483)	0.304 (5.454)
Transaction (share of parcel-years)	0.0402 (0.1964)	0.0419 (0.2004)	0.0458 (0.2090)	0.0455 (0.2084)
Foreclosure filing (share)	0.0107 (0.1027)	0.0095 (0.0971)	0.0064 (0.0795)	0.0037 (0.0608)
Distressed sale (share)	0.0022 (0.0465)	0.0017 (0.0416)	0.0012 (0.0343)	0.0007 (0.0259)
Year built	1986 (19)	1993 (14)	1993 (15)	1992 (17)
Living area (sq. ft.)	1,519 (644)	1,768 (472)	2,306 (586)	3,351 (1,001)
Bedrooms	2.94 (0.58)	3.11 (0.50)	3.45 (0.65)	3.94 (0.70)
Bathrooms	2.07 (0.62)	2.33 (0.41)	2.51 (0.46)	3.14 (0.87)
Parcel-years	297,023	316,813	332,832	342,399
Parcels	33,048	33,568	33,664	33,633
<i>Panel B. Transfers, 2011–2022</i>				
Sale price (2025\$)	244,612 (183,251)	271,780 (145,927)	371,450 (170,802)	694,142 (414,005)
Price per sq. ft. (2025\$)	157.27 (69.96)	156.67 (58.37)	167.78 (63.07)	205.89 (87.84)
Sales	12,955	14,094	15,938	16,056

Notes. The triple-difference estimation sample: parcel-years over $\tau \in [-3, +4]$ around the 2011 and 2019 Mecklenburg assessments, stacked across both events. Cells report means by value quartile with standard deviations in parentheses beneath, cut within county so that Q1 is the cheapest quarter of that county's own stock. Quartiles are assigned from each parcel's most recent arm's-length sale price between 1998 and 2011, expressed in 2025 dollars. A transaction is any priced transfer, arm's-length or otherwise; the sale price and price per square foot are conditional on transacting. A foreclosure filing is the onset of the foreclosure process and conveys no title. Assessed values, millage and the tax bill are reported alongside the outcomes because they exist in administrative form for Mecklenburg, which is what makes the tax shock measurable rather than inferred. Continuous variables are winsorized at the 1st and 99th percentiles.

4.1 Deed coverage and selection into the estimation sample

A parcel enters the estimation sample only if it recorded an arm's-length sale between 1998 and 2011, since those sales are what place it in the value distribution. This requirement is innocuous where deed recording is complete and corrosive where it is not. If a county's recorded transfers are sparse, then clearing the requirement identifies a parcel that happens to be well documented rather than one that is typical, and being well documented is itself correlated with transacting. The sample would then be selected on precisely the margin the transaction outcomes are meant to measure.

I test this directly. Within each county's owner-occupied residential stock, I compare the later transfer rate of parcels carrying a qualifying sale against those that do not. Where recording is adequate the two rates are indistinguishable, which is what the requirement should produce if it is only a measurement device and not a filter on behavior. Where recording is sparse, the parcels that clear it go on to transact four to seven times as often as those that do not, and the requirement is no longer measuring anything about housing markets.

The two cases separate sharply on how many transfer deeds a county records, and nothing falls between them. Counties recording at least roughly four transfer deeds for every ten parcels across the window all pass the comparison; those recording fewer than three all fail it. I therefore retain counties above that threshold, which admits every county whose sample survives the neutrality test and excludes every county that does not. Three counties fall below it and are set aside. The threshold is a data-quality screen, chosen after inspecting the diagnostic rather than in advance of it, and it is applied to the outcome-free question of whether a county's deeds are recorded densely enough for its sample to be representative of its own stock.

Two features of the retained sample are worth noting because they are easy to mistake for the same problem. A county can anchor a small share of its stock without being selected: where owner-occupancy is identified through a homestead exemption, the flagged owners are longer-tenured and therefore less likely to have sold inside the window, which shrinks the anchored share without distorting who is in it. Conversely, a high anchoring share is no guarantee of neutrality on its own. The comparison of transfer rates, not the anchoring share, is what separates the two.

4.2 Which counties build the comparison unit

The donor pool admits thirty-three counties, but the estimator does not use them equally. Weights are fitted separately for every outcome, assessment and side of the contrast, so each cell of the results has its own comparison group, and Tables 2 and 3 report which counties those are.

Three features govern how the estimates should be read. The weights are concentrated: the 2011 cells

draw on four to eleven donors each, with the largest single county carrying between a quarter and half of the fit, so several estimates rest substantially on one or two comparison counties. The composition turns over between assessments: twenty donors receive weight somewhere in 2011 and twenty-nine in 2019, with seventeen common to both, which follows from solving (3) cell by cell rather than imposing one weight vector across outcomes. And participation is broad even where weight is not: only one county in the pool receives no weight in any cell, so the concentration reflects which counties fit best rather than a pool most of whose members are unusable.

Two consequences follow. Where an estimate rests mostly on one county, it is a different claim from one resting on many, and the tables let a reader see which they are looking at. And where two cells point the same way, they do so from separately chosen comparison groups, so the agreement is corroboration rather than a shared construction reproducing itself.

Table 2: Donor composition of synthetic Mecklenburg, by outcome and assessment.

Donor	Log sale price 2011	Log sale price 2019	Log price/sq. ft. 2011	Log price/sq. ft. 2019	Foreclosure filing 2011	Transaction 2011	Transaction 2019	Distressed sale 2011	Mean
Cobb GA	33.6	—	—	—	33.4	26.2	—	—	11.6
Shelby AL	43.0	—	5.2	8.9	32.9	—	—	—	11.2
Chesterfield VA	—	—	53.3	—	—	34.4	1.3	—	11.1
Hampton VA	1.8	—	12.6	—	—	28.5	43.1	—	10.7
Newport News VA	15.4	—	10.1	—	17.0	—	—	—	5.3
Manatee FL	—	12.8	6.0	—	—	—	—	21.4	5.0
Ramsey MN	—	2.8	—	—	—	—	37.1	—	5.0
Polk FL	1.3	11.5	—	2.7	—	10.9	—	11.4	4.7
Hillsborough FL	—	5.6	—	31.6	—	—	—	—	4.7
Oakland MI	—	8.2	—	—	16.7	—	—	9.3	4.3
Sarasota FL	—	—	—	—	—	—	—	25.2	3.2
Duval FL	—	7.6	—	13.8	—	—	—	—	2.7
Pasco FL	—	14.9	—	5.5	—	—	—	—	2.6
Broward FL	—	—	—	19.7	—	—	—	—	2.5
Lee FL	—	—	—	—	—	—	17.9	—	2.2
Fairfax VA	5.0	—	12.2	—	—	—	—	—	2.1
Washington MN	—	8.4	—	8.5	—	—	—	—	2.1
Osceola FL	—	15.7	—	—	—	—	—	—	2.0
Collier FL	—	—	—	—	—	—	—	14.3	1.8
Brevard FL	—	—	—	—	—	—	—	9.8	1.2
Orange FL	—	6.2	—	1.9	—	—	—	—	1.0
Pima AZ	—	—	0.6	6.2	—	—	—	—	0.8
Loudoun VA	—	—	—	—	—	—	—	6.7	0.8
Scott MN	—	5.7	—	—	—	—	—	—	0.7
Prince William VA	—	—	—	—	—	—	—	1.8	0.2
Jefferson AL	—	—	—	1.1	—	—	—	—	0.1
Lake FL	—	—	—	—	—	—	0.6	—	0.1
<i>6 further donors</i>	—	—	—	—	—	—	—	—	—

Notes. Weights in percent, from (3), for the cells that pass the feasibility screen; cells the estimator cannot fit are omitted because a composition without a corresponding estimate invites over-reading. Weights below half a percent are shown as dashes, the optimizer returning small positive values rather than exact zeros. The final row collects donors receiving no weight in any cell; they are counted rather than dropped, since the share of the pool that contributes nothing is itself a measure of how concentrated the comparison is. The mean column averages across the five cells and is a summary of participation, not a weight used in any estimate.

Table 3: Donor composition for quartiles two to four.

Donor	Log sale price 2011	Log sale price 2019	Log price/sq. ft. 2011	Log price/sq. ft. 2019	Foreclosure filing 2011	Transaction 2011	Transaction 2019	Distressed sale 2011	Mean
Chesterfield VA	53.6	—	47.6	—	—	51.9	—	—	19.1
Shelby AL	14.9	3.8	—	12.9	33.9	—	14.0	—	9.9
Oakland MI	—	2.8	—	10.0	35.6	—	23.4	—	9.0
Fairfax VA	19.2	16.1	27.5	4.8	—	—	—	—	8.5
Cobb GA	10.5	—	13.0	—	21.0	16.9	—	3.8	8.1
Newport News VA	—	4.8	11.9	4.3	1.5	13.6	16.7	4.1	7.1
Polk FL	—	20.3	—	15.1	—	—	—	5.8	5.1
Hampton VA	1.8	4.5	—	5.9	8.0	17.6	—	—	4.7
Brevard FL	—	6.3	—	3.0	—	—	—	27.8	4.6
Manatee FL	—	—	—	4.2	—	—	—	26.5	3.8
Ramsey MN	—	10.4	—	11.0	—	—	5.6	—	3.4
Osceola FL	—	—	—	5.7	—	—	—	12.3	2.3
Miami-Dade FL	—	3.6	—	—	—	—	10.3	3.4	2.2
Charlotte FL	—	8.8	—	7.4	—	—	—	—	2.0
Scott MN	—	8.8	—	6.1	—	—	—	—	1.9
Pima AZ	—	—	—	—	—	—	14.4	—	1.8
Marion FL	—	—	—	—	—	—	10.0	—	1.2
Jefferson AL	—	5.0	—	4.8	—	—	—	—	1.2
Pasco FL	—	—	—	—	—	—	—	8.8	1.1
Lake FL	—	—	—	3.0	—	—	—	2.1	0.6
Duval FL	—	4.9	—	—	—	—	—	—	0.6
Prince William VA	—	—	—	—	—	—	—	4.0	0.5
Collier FL	—	—	—	—	—	—	3.1	—	0.4
Loudoun VA	—	—	—	—	—	—	2.6	—	0.3
Orange FL	—	—	—	—	—	—	—	1.5	0.2
Dakota MN	—	—	—	1.4	—	—	—	—	0.2
<i>7 further donors</i>	—	—	—	—	—	—	—	—	—

Notes. As Table 2, for the pooled remainder of the distribution rather than the cheapest quartile. Weights are fitted separately for the two sides of the contrast, so the counties reproducing Mecklenburg's bottom quartile need not be those reproducing the rest of its stock, and in several cells they are not.

4.3 Regressivity

I fit a trendline on sales price and assement value to characterize regressivity in assessment rates in Figure 1. This figure makes the same point in the cross-section: the assessor's value rises less than one-for-one with a sale-based market value, so the assessment ratio is regressive across the value distribution. Each data point represent a parcel year observation using the same weightes sales price measure for market value. The 45-degree line (dashed) represents as assessment rate of one. Notably, very few parcels above the median market value sit above the 45-degree line.

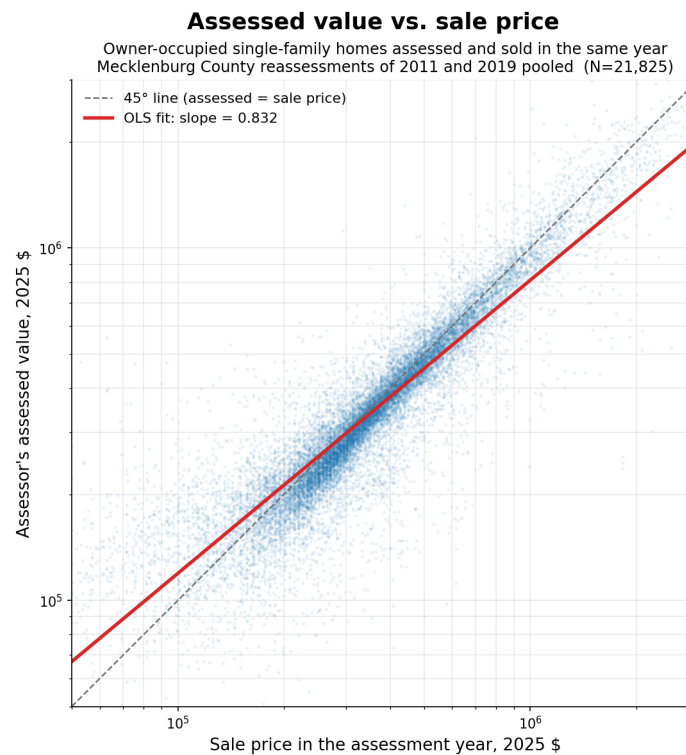


Figure 1: Assessed vs. sale-based value, Mecklenburg County for owner-occupied homes. The assessor's value against a sale-anchored market value; the below-one slope is the regressivity of the assessment.

Next, I document variation in the annual change in the property tax bill across home-value quartiles. This measure is inclusive of all property tax components, such as homestead exemptions and millage rate differences, because it is measured from the final bill the owner owes. Table 4 displays this variation for the four quartiles of Mecklenburg homes.

Table 4: Change in the property tax bill (%) by home-value quartile, Mecklenburg, 2012–2022.

	Millage (per \$100)	Change in tax bill (%)			
		Q1 (cheapest)	Q2	Q3	Q4 (most expensive)
2012	1.177	-2.33	-2.28	-2.18	-2.17
2013	1.206	1.99	2.03	2.03	1.69
2014	1.227	0.24	1.63	1.74	1.76
2015	1.226	0.78	0.69	0.64	0.68
2016	1.235	0.19	0.14	0.18	0.31
2017	1.236	0.23	0.10	0.18	0.24
2018	1.251	1.29	1.27	1.24	1.20
2019	0.974	24.14	19.42	12.42	3.70
2020	0.981	1.27	0.86	0.68	0.60
2021	0.981	-0.75	-0.10	0.51	1.05
2022	0.929	-4.10	-4.29	-4.03	-3.84

Notes. Mean year-over-year change in the property tax bill, in percent, for owner-occupied Mecklenburg homes by value quartile (Q1 cheapest). The bill is the county’s administrative levy, and the change is nominal: the bill is a nominal claim on the owner’s cash, and inflation is common to all parcels within a year, so it shifts every quartile equally and cancels out of the comparison across quartiles. Quartiles are cut on each home’s most recent arm’s-length sale price between 1998 and 2011, expressed in 2025 dollars. Millage is the combined county and municipal rate per \$100 of assessed value. The window opens in 2012 because the change requires a prior year. The horizontal rule marks the 2019 countywide assessment; the prior cycle began in 2011. Between assessments the bill still moves, because the millage rate is set annually even though the assessed value is frozen: the county-wide change runs from -4.9 to $+2.0$ percent across those years. What the assessment adds is not movement but a gradient, the change in a reappraisal year running from about 24 percent in the cheapest quartile to 4 percent in the most expensive, where in every other year the four quartiles move within a fraction of a point of one another. Changes are winsorized at the 1st and 99th percentiles.

5 Results

5.1 Triple-Difference Estimates

Table 5 reports the synthetic control for the 2011 assessment, which is the only event with a full and symmetric window on either side of it, the transaction extract ending in 2020 and leaving the 2019 assessment two post-assessment years against four before it.

All five outcomes can be estimated. Four agree in sign with the triple difference: sale prices, prices per square foot and transaction probability fall for the cheapest homes, and foreclosure filings rise. Foreclosure filings rise by 0.20 percentage points, the only estimate that holds its sign across every donor pool and both frequencies. The price estimates do not survive the move to quarterly cells: at annual frequency the cheapest homes’ prices fall by roughly 2%, and at quarterly frequency, where sixteen pre-assessment points replace four, the estimate falls to 0.8% and the pre-assessment fit degrades fivefold. While none is significant

against the donor placebo distribution—whose thirty-three donors yield an attainable p -value of 0.03—the filing estimate reaches 0.24.

The two designs are not the same specification, and the magnitudes should be read with that in mind. The triple difference is a regression on parcel-years carrying hedonic controls for bedrooms, bathrooms, living area and year built, with county by event fixed effects and standard errors clustered by parcel. The synthetic control runs on county-year-quartile means and carries no hedonic controls at all: subtracting each county’s own pre-assessment mean does the work of a unit fixed effect, taking the cheapest quartile against the rest removes whatever moved that county as a whole in a given year, the donor weights stand in for covariate adjustment, and inference comes from the placebo distribution rather than from a clustered variance. What this leaves unaddressed is composition. If the mix of homes transacting within a quartile shifts differently across counties, the triple difference absorbs part of that shift and the synthetic control absorbs none of it, which is one reason to carry price per square foot alongside the price itself, since dividing by living area is the only composition adjustment the synthetic control makes and it is mechanical rather than estimated. The agreement in sign is therefore evidence from a second estimator with different assumptions, not a second measurement of the same quantity, and the gap between the two price estimates should not be read as though both numbers were estimating the same conditional mean. The absence of covariate adjustment is a property of the estimator rather than an omission: donor weights are chosen to reproduce Mecklenburg’s pre-assessment path across the whole value distribution, which is what a synthetic control offers in place of controls, and adjusting the outcomes for structure characteristics before aggregating would compress the very contrast between the cheapest quartile and the rest that the design identifies from, those characteristics being close proxies for value.⁸ The comparison with the triple difference is therefore not exact, but it is informative for that reason rather than in spite of it, the two estimators sharing neither a control group nor an identifying assumption.

The agreement is more informative than three numbers pointing the same way would usually be, because the three cells are built from different counties. The price cells rest on Chesterfield for two-thirds or more of their weight, while the filing cell rests on Hennepin, with Chesterfield contributing a seventh; the composition is in Table 2. A single well-matched county driving all three would be one comparison reported three times, and that is not what produces this.

Figure 2 shows the paths behind those numbers. The price effects are concentrated in the first two years after the assessment rather than persisting, and the filing effect appears late: Mecklenburg and its synthetic

⁸Regressing each outcome on bedrooms, bathrooms, living area and year built within county and year, and aggregating the residuals, reduces Mecklenburg’s pre-assessment gap between the cheapest quartile and the rest from -0.746 to -0.207 in log sale price, from 0.510 to 0.263 percentage points in foreclosure filings, and from 0.985 to 0.496 percentage points in completed distressed sales. An adjustment made within quartiles rather than across them would leave the contrast intact, and whether it materially changes the estimates is left open here.

counterpart rise together through $\tau = +1$ and separate only afterwards, which a single post-period average conceals.

Contributing donor counties, 2011 assessment (legend: weight on cheapest quartile / on the rest, %)

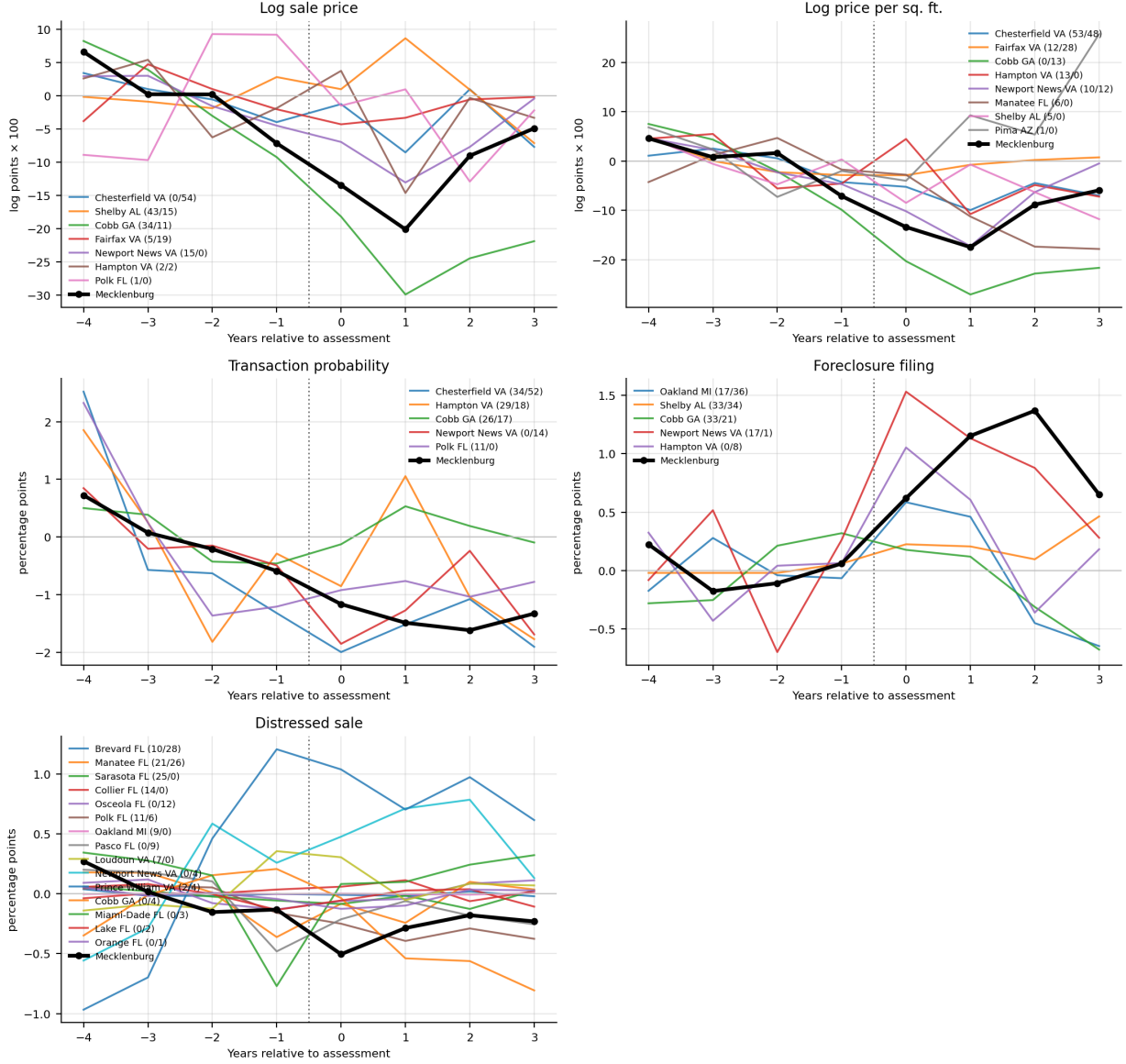


Figure 3: The contributing donor counties, 2011 assessment.

Notes. Each county carrying weight in a cell, drawn on the same contrast as Figure 2: the cheapest quartile minus the mean of the other three, demeaned on the pre-assessment window. Mecklenburg is heavy black. Legend entries give each donor's weight on the cheapest quartile and on the pooled remainder respectively, in percent, since the two sides are fitted separately and a county can enter one and not the other. Counties carrying less than half a percent on both sides are omitted; the optimizer returns small positive values rather than exact zeros. This figure exists because a weighted average can be well behaved while its components move in opposite directions and cancel, which the synthetic line alone cannot reveal.

Mecklenburg against its synthetic control, 2011 reassessment

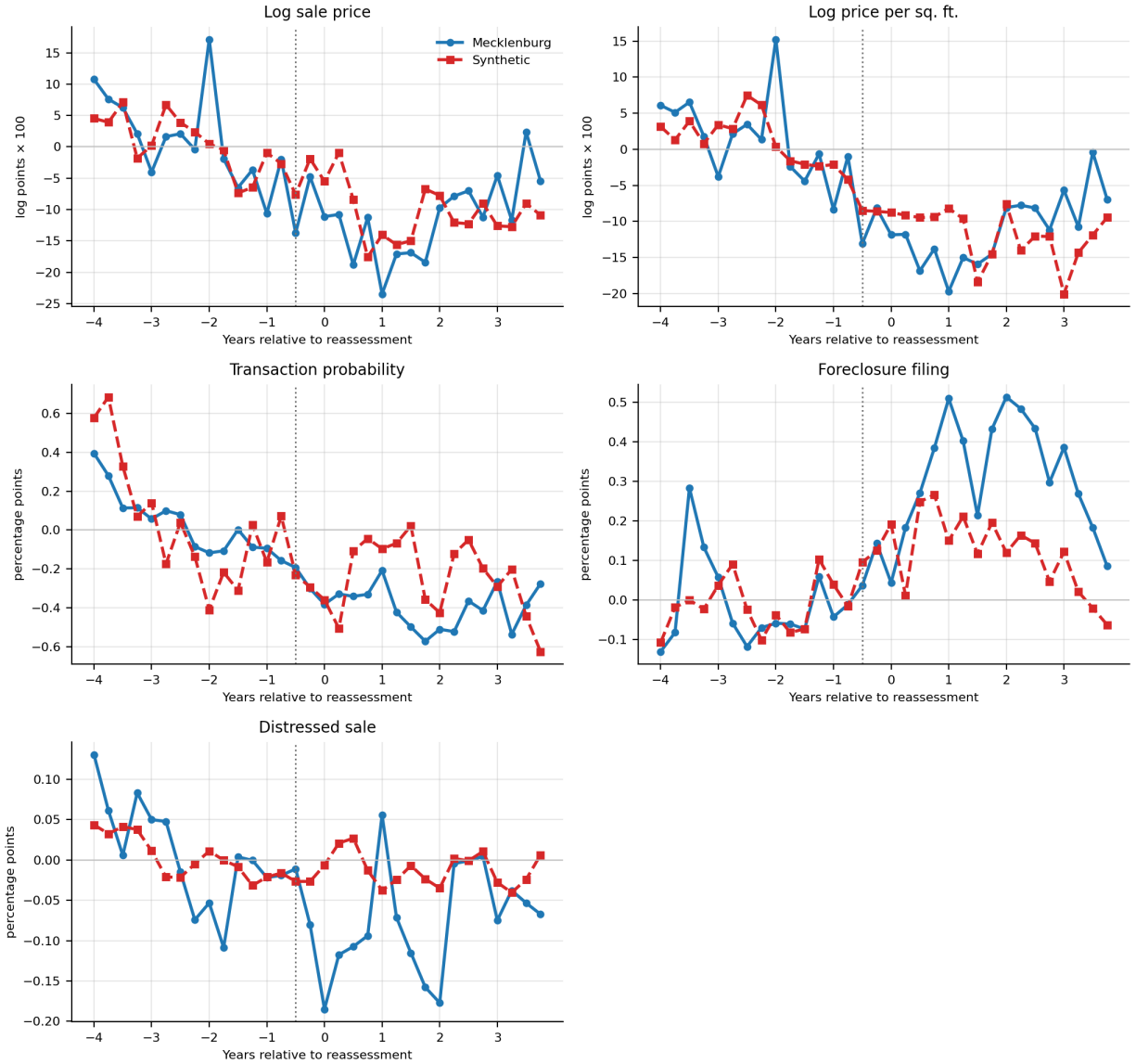


Figure 2: Mecklenburg against its synthetic control, 2011 assessment.

Notes. The contrast the design identifies from: the cheapest quartile minus the mean of the other three, each side demeaned on its own pre-assessment window. Solid is Mecklenburg, dashed the weighted donors; because weights are fitted separately for the cheapest quartile and for the remainder, the dashed line is itself a difference of two separately constructed synthetic counties. The vertical rule marks the assessment. Agreement to its left is what the weights were fitted to deliver and is not evidence that the design works; the figure is a diagnostic for the shape of the post-assessment separation, and for whether a fit was achievable at all. Only cells passing the feasibility screen are drawn, and the pre-assessment fit is visibly tighter for foreclosure filings than for the price outcomes, matching their fit ratios.

Table 5: Synthetic control estimates, 2011 assessment.

Outcome	Effect	Pre-period fit	Rank	p
Log sale price	-0.008	0.0587	25/29	0.86
Log price/sq. ft.	0.007	0.0515	24/29	0.83
Foreclosure filing	0.198 pp	0.0010	7/29	0.24
Transaction	-0.156 pp	0.0019	22/29	0.76
Distressed sale	-0.064 pp	0.0005	21/29	0.72

Notes. Estimates of (4) on the 2011 assessment, over $\tau = -4$ to -1 before and $\tau = 0$ to $+3$ after. Price effects are in log points, the filing effect in percentage points. Only the 2011 event is reported: the transaction extract ends in 2020, so the 2019 assessment has a two-year post-period against a four-year pre-period, and its deed-based outcomes are measured over a window too short and too asymmetric to set beside 2011. Pre-period fit is the root mean squared discrepancy between the treated county’s demeaned gap and its synthetic counterpart before the assessment, in the units of the outcome. Inference is by permutation on the ratio of post-period to pre-period fit, following the standard practice for this estimator: a unit that was never tracked closely will post a large post-period gap whether or not anything happened to it, and normalising by the pre-period fit is what separates the two cases. Rank is the treated county’s position among itself and the eligible donors on that ratio, and p inverts the rank on the ratio of post-period to pre-period fit. Eligibility is decided separately for each assessment, so the number of units differs between the two and is reported in the table.

5.2 IV Results

Table 6 reports the first stage and the reduced form. The instrument moves the bill by 25.6 percentage points in the cheapest quartile and by 3.5 in the most expensive, and the reduced form inherits that ordering: a reappraisal raises the probability of a foreclosure filing by 0.75 percentage points among the cheapest homes and by 0.13 among the most expensive, and it lowers their sale prices by about 15 percent while leaving the most expensive quartile’s prices higher. Both rows are ordinary regressions on the instrument, so they are valid at every point of the distribution, whatever the strength of the first stage.

Table 6: First stage and reduced form by home-value quartile.

	Q1 (cheapest)	Q2	Q3	Q4 (most expensive)
<i>First stage: percent change in the tax bill</i>				
Reassessment year	25.55 (1.51)	20.81 (1.65)	13.21 (1.76)	3.47 (1.24)
Conditional F , sale-price sample	211	174	55	4
Conditional F , filing sample	286	159	56	8
<i>Reduced form: effect of a reappraisal</i>				
Log sale price (%)	-14.97 (3.15)	-13.25 (2.53)	-1.96 (3.11)	14.76 (4.01)
Foreclosure filing (pp)	0.749 (0.068)	0.628 (0.089)	0.372 (0.054)	0.133 (0.064)
Observations (filing)	127,929	134,121	126,353	141,976

Notes. Owner-occupied Mecklenburg parcels, 2012–2023. The first stage regresses the percent change in the administrative tax bill on an indicator for a reappraisal year within each home-value quartile; the reduced form regresses each outcome on the same indicator. Controls, fixed effects, and clustering as in equation (5). The conditional F is the Sanderson–Windmeijer statistic, which within a quartile is the squared t on the instrument. The sale-price sample is restricted to transacting parcel-years and is smaller.

6 Conclusion

Two designs built on the same statutory assessment cycle point the same way on distress. The triple difference, which sets the assessing county against a weighted combination of annually assessing counties, shows that an assessment raises foreclosure filings among the cheapest homes by 0.20 percentage points, an effect that holds its sign across every donor pool and frequency the data support, though it is not unusual against the distribution of placebo effects. That design finds no price effect once the pre-assessment window carries enough resolution to fit honestly: at quarterly frequency, where sixteen pre-assessment observations replace four, the estimated price decline falls to under a percentage point.

The instrumental-variables estimates, which replace the assessment indicator with the tax bill it produces, show why distress rather than price is where the burden surfaces. Measured per point of tax the response is statistically indistinguishable across the quartiles in which it can be measured at all; what differs is the tax. Reappraisal raises the cheapest quartile's bill by 25.6 percentage points against 3.5 for the most expensive, and the foreclosure increase that follows is 0.75 percentage points against 0.13. The regressive burden of over-assessment therefore reaches low-value owners through the size of the correction rather than through any greater sensitivity to it, which makes the mechanism a property of the assessment system rather than of the households it catches.

Two limits bound what these estimates establish. The synthetic control identifies the effect from a single reappraisal in one county, and its placebo distribution places the filing estimate seventh of thirty-three, short of conventional significance; the design constrains the number of comparison units, and no expansion of the donor pool changed that. And the two designs disagree on prices, the instrumental-variables estimates finding a graded discount that the synthetic control does not detect. What survives both is the distributional claim: the correction is larger for cheaper homes, and the distress that follows is larger with it. The result identifies a specific cost of a widely used assessment system and a margin—foreclosure among the cheapest homes—where reform would bind.

A Appendix

A.1 Synthetic control paths at the 2019 assessment

The 2019 assessment is not estimated in the main text, because the transaction extract ends in 2020 and leaves it two post-assessment years against four before it. Figure A1 shows the two outcomes that nonetheless pass the feasibility screen there, so that the excluded event is visible rather than merely asserted to be unusable. The pre-assessment fit is closer than at 2011, which is what six pre-period years rather than four buys; the post-period is too short to read a shape from, which is why it is excluded.

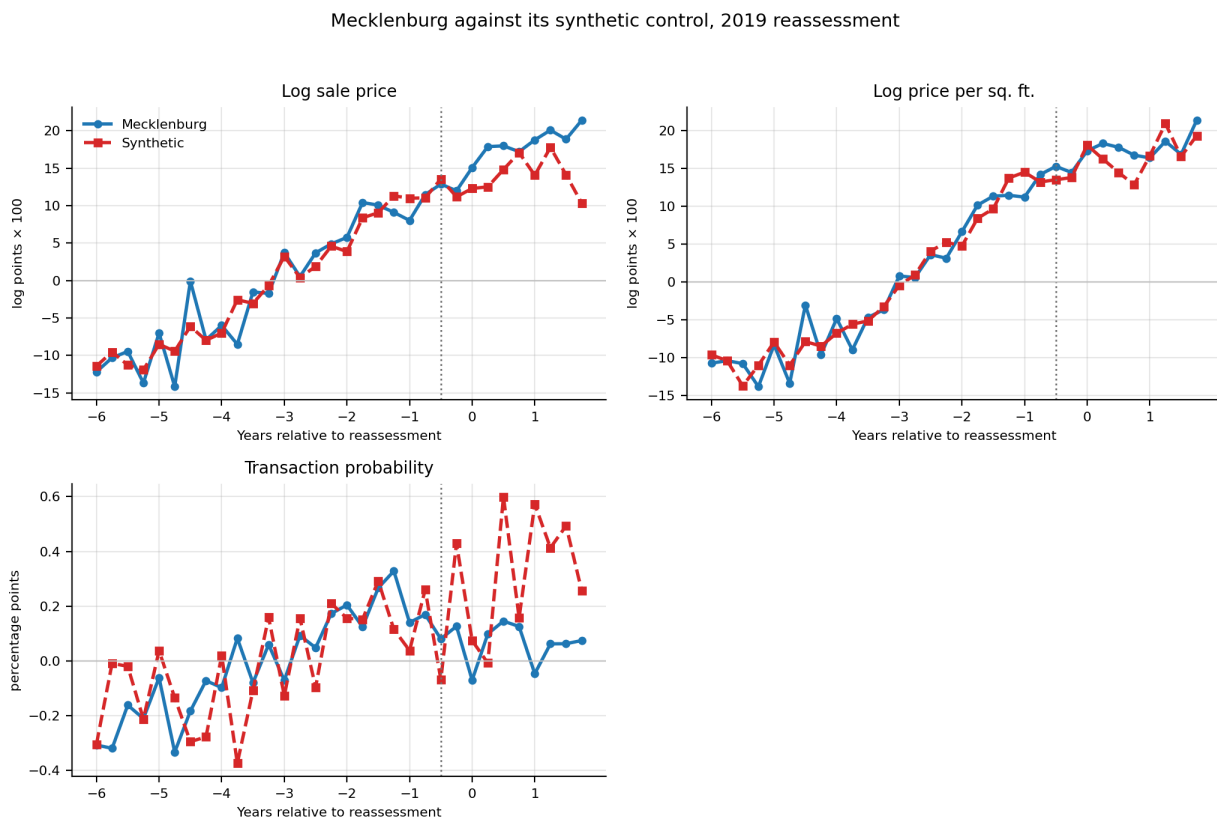


Figure A1: Mecklenburg against its synthetic control, 2019 assessment.

Notes. As Figure 2, for the 2019 assessment, over $\tau = -6$ to -1 before and $\tau = 0$ to $+1$ after. Only log sale price and log price per square foot pass the feasibility screen at this event.

A.2 Industry measures of assessment performance

The appraisal industry evaluates assessment quality with a standard set of statistics on the distribution of the assessment ratio, the assessor's value divided by the sale price. Table A1 reports them for Mecklenburg following McMillen and Singh (2020). The value-weighted mean weights each ratio by its sale price, so it

Contributing donor counties, 2019 assessment (legend: weight on cheapest quartile / on the rest, %)

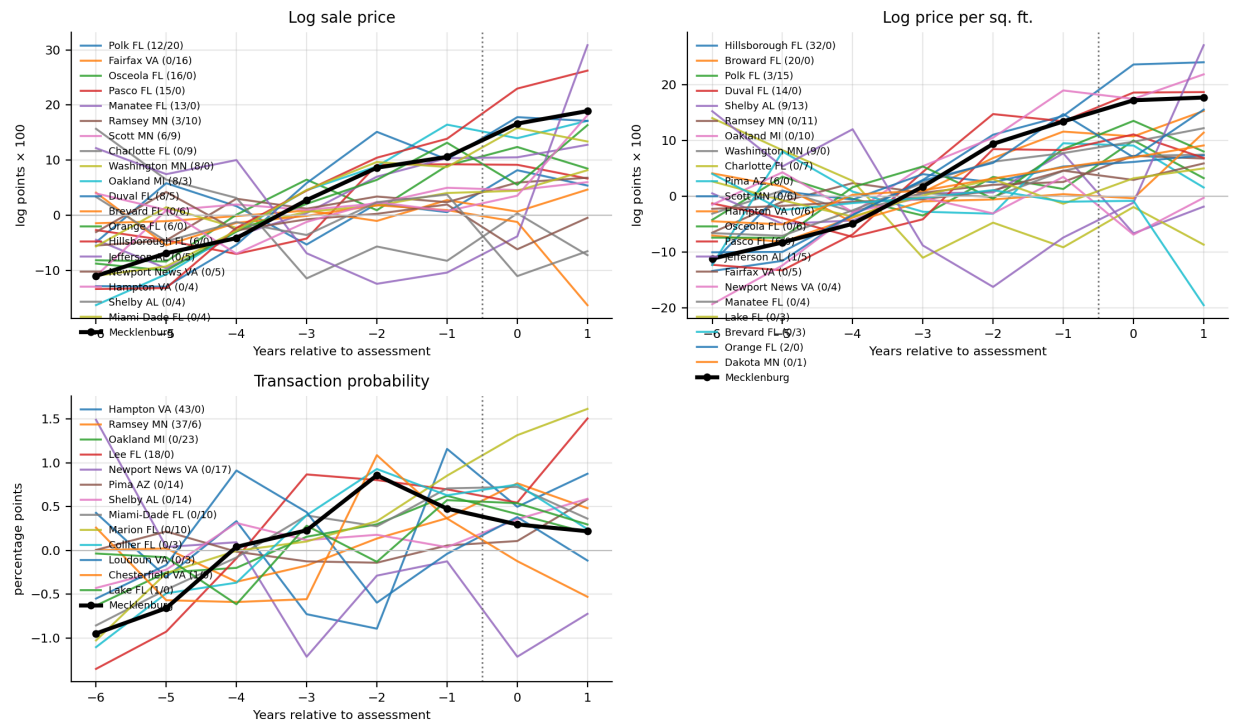


Figure A2: The contributing donor counties, 2019 assessment.

Notes. As Figure 3, for the 2019 assessment. More counties carry weight at this event than at 2011, the longer pre-period admitting a wider set.

equals total assessed value divided by total sale price; when it falls below the arithmetic mean, expensive homes are being assessed at lower ratios. The price-related differential is the ratio of the two, and values above one indicate regressivity. The coefficient of dispersion measures uniformity as the average absolute deviation from the median ratio.

By these measures Mecklenburg appears close to compliant. The price-related differential is 1.022 in 2011 and 1.010 in 2019, both inside the industry’s vertical-equity range of 0.98 to 1.03, and both coefficients of dispersion fall below the uniformity threshold of 15. Yet the same homes yield fitted slopes well below one. The two verdicts differ because the price-related differential is a ratio of two aggregates and is insensitive to curvature in the ratio distribution. Figure A3 shows the curvature directly: binned mean ratios fall from roughly 1.20 to 1.02 across the 2011 price range, while the price-weighted aggregate barely moves because most assessed value sits in the middle of the distribution, where ratios are near one. The regressivity in Mecklenburg is therefore of a kind the standard industry diagnostic does not register.

Table A1: Assessment ratio statistics, Mecklenburg assessments of 2011 and 2019.

Reassessment	<i>N</i>	Central tendency			Dispersion		PRD
		Median	Mean	Value-weighted	COD	Value-weighted	
2011	5,230	1.027	1.046	1.024	13.2	12.8	1.022
2019	15,563	0.931	0.925	0.916	11.3	11.4	1.010
Pooled	20,793	0.948	0.955	0.938	12.4	12.0	1.019

Notes. Owner-occupied single-family Mecklenburg homes assessed and sold in the same year. The assessment ratio is the assessor’s value divided by the sale price, so it is unit free and needs no price adjustment. The value-weighted mean weights each ratio by its sale price; the value-weighted coefficient of dispersion (COD) weights the absolute deviations the same way. The price-related differential (PRD) is the arithmetic mean divided by the value-weighted mean, with values above one indicating that expensive homes are assessed at lower ratios. Industry standards call for a COD below 15 and a PRD between 0.98 and 1.03. Outliers are removed with the nonparametric trim used by McMillen and Singh (2020), which drops observations outside three interquartile ranges of the first and third quartiles of the log sale price, the log assessed value, and the assessment ratio.

A.3 Assessed value against sale price by assessment year

Figure 1 pools the 2011 and 2019 assessments. Figure A4 splits them, and the two cycles differ: the fitted slope is 0.777 in 2011 and 0.905 in 2019, so assessments tracked sale prices more closely in the later cycle even though both remain below one.

A.4 Effective tax rates

Figure A5 shows how the effective tax rate is progressively lower than market value for each quartile in 2019 and then drifts from market value with time from assessment for all quartiles.

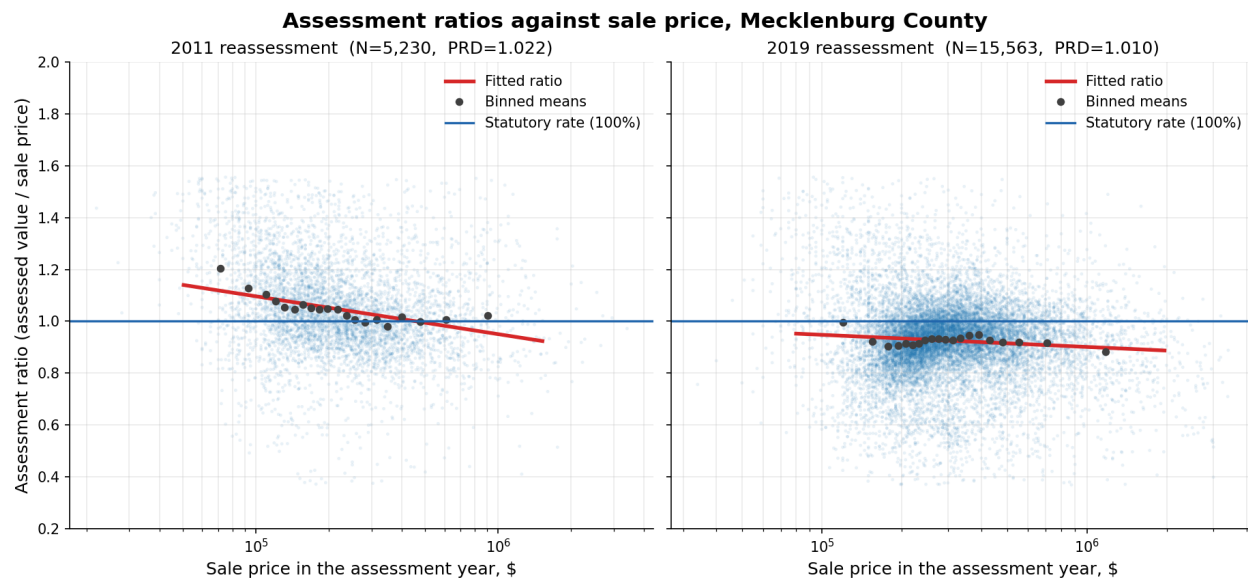


Figure A3: Assessment ratios against sale price, owner-occupied single-family Mecklenburg homes assessed and sold in the same year. The horizontal line marks North Carolina’s statutory assessment rate of 100 percent of market value. The fitted line is a linear regression of the ratio on the log sale price; the binned means are averages within twenty equal-count bins of the sale price and trace the curvature the linear fit averages over.

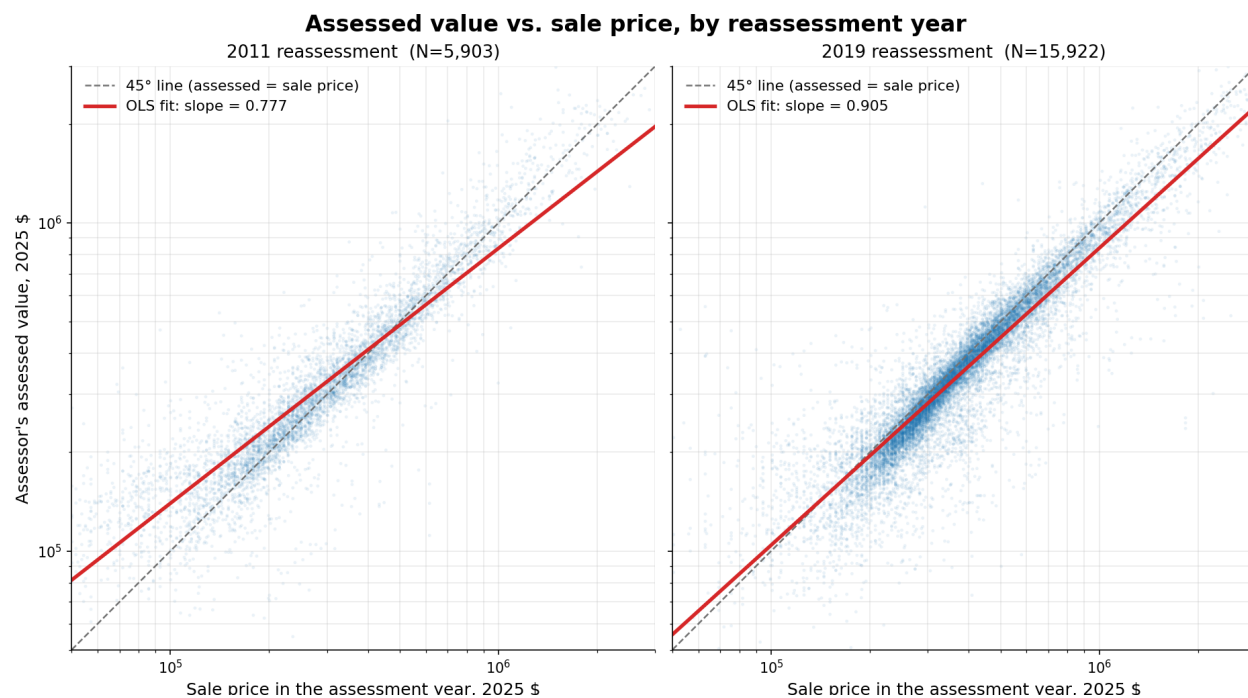


Figure A4: Assessed value against sale price by assessment year, owner-occupied single-family Mecklenburg homes assessed and sold in the same year. Both axes are in 2025 dollars. The dashed line marks equality between the assessed value and the sale price; the fitted slope lies below one in each cycle, which is the regressivity of the assessment.

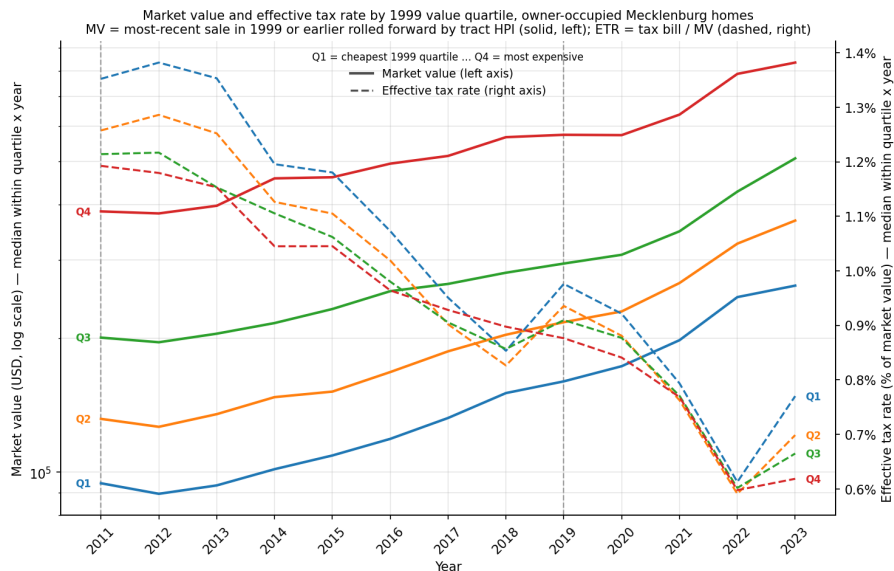


Figure A5: Annual market values and effective tax rates in Mecklenburg County for owner-occupied homes. Effective tax rates (dashed) and market value (solid) by 1999 home-value quartile.

A.5 Regressivity and Imputed Market Values

I document regressivity in both assessment rates and effective tax rates with respect to sales prices. Figure A6 shows the divergence between the log of assessment values (dashed) and the log of sales prices (solid) over time for 1999-value quartiles in Mecklenburg. At assessment years 2011 and 2019, the two measures track closely and then diverge increasingly with time. Assessment rates are clearly largest for the cheapest homes and fall progressively further below it for more expensive ones in both assessment years and subsequent years. Aside from a county-wide correction to 2011 assessed values in 2014, assessment values are frozen nonassessment years.

The assessment reset is where the regressivity compounds. Between 2018 and 2019 the effective rate on the cheapest quartile rises by roughly 15% (from 0.98 to 1.13), while the most expensive quartile's is essentially unchanged—a decline of about 1%—and the second and third quartiles fall monotonically in between, near 11% and 9%. The reappraisal, nominally a uniform return to full-value assessment, thus raises the tax burden almost entirely on low-value homes: the cheapest owners carry the highest effective rate throughout the cycle and then absorb nearly all of its upward reset, while the most expensive homes are left untouched. This asymmetric reset at the bottom of the value distribution is the compounding that motivates estimating the distress and price effects separately by home value.

Table 4 reports a single millage rate per year, since the rate is set at the jurisdiction level rather than by home value. Table A2 breaks the rate out by quartile. The four series are nearly identical within each year,

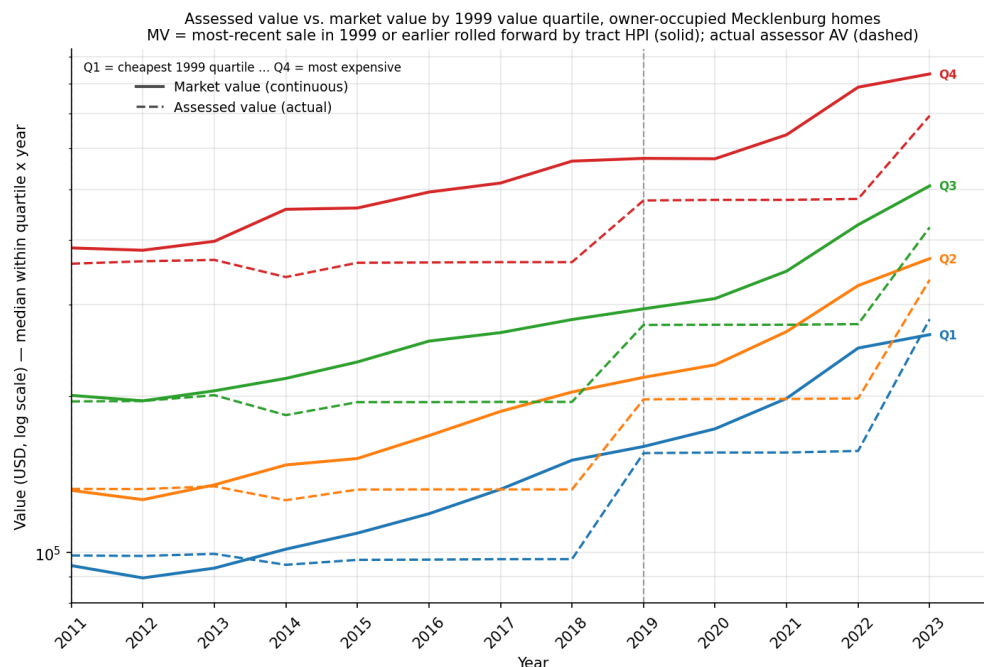


Figure A6: Assessment gaps in Mecklenburg County for owner-occupied homes. Median assessed value (dashed) and imputed market value (solid) by 1999 home-value quartile; the assessed-value shortfall widens with value and time from assessment.

so the gap in tax-bill growth across quartiles reflects the assessment rather than any difference in statutory rates.

Table A2: Millage rate and change in the property tax bill by home-value quartile, Mecklenburg, 2012–2022.

	Millage (per \$100)				Change in tax bill (%)			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
2012	1.190	1.182	1.166	1.171	-2.33	-2.28	-2.18	-2.17
2013	1.211	1.201	1.197	1.213	1.99	2.03	2.03	1.69
2014	1.232	1.224	1.216	1.234	0.24	1.63	1.74	1.76
2015	1.244	1.239	1.213	1.208	0.78	0.69	0.64	0.68
2016	1.249	1.242	1.224	1.229	0.19	0.14	0.18	0.31
2017	1.250	1.242	1.225	1.229	0.23	0.10	0.18	0.24
2018	1.266	1.258	1.240	1.244	1.29	1.27	1.24	1.20
2019	0.984	0.979	0.967	0.967	24.14	19.42	12.42	3.70
2020	0.992	0.986	0.973	0.974	1.27	0.86	0.68	0.60
2021	0.992	0.986	0.973	0.974	-0.75	-0.10	0.51	1.05
2022	0.936	0.933	0.923	0.926	-4.10	-4.29	-4.03	-3.84

Notes. Owner-occupied Mecklenburg parcels by value quartile (Q1 cheapest, Q4 most expensive). Millage is the combined county and municipal rate per \$100 of assessed value, averaged across parcels within the quartile; parcels in different municipalities face different combined rates, so the quartile means differ only through municipal composition. The change in the tax bill is defined in equation (7). The horizontal rule marks the 2019 countywide assessment.

where the change in the property tax bill is

$$\Delta T_{it} = 100 \times \left(\frac{T_{it}}{T_{i,t-1}} - 1 \right), \quad (7)$$

with T_{it} the administrative tax levy on parcel i in year t . Dollar levels reported elsewhere are expressed in 2025 dollars, but the change is left in nominal terms because inflation is common to all parcels within a year and therefore drops out of any comparison across quartiles or out of any specification carrying year fixed effects.

References

- Abadie, Alberto, Alexis Diamond, and Jens Hainmueller**, “Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California’s Tobacco Control Program,” *Journal of the American Statistical Association*, 2010, *105* (490), 493–505.
- Amornsiripanitch, Natee**, “Why Are Residential Property Tax Rates Regressive?,” Working Paper, Federal Reserve Bank of Philadelphia November 2020. Finalist, 2020 BlackRock Applied Research Award; 2021 Outstanding Doctoral Dissertation in Government Finance and Taxation.
- , “Why Are Residential Property Tax Rates Regressive?,” Working Paper, Federal Reserve Bank of Philadelphia April 2024. Revised version of Amornsiripanitch (2020); dated April 19, 2024.
- Avenancio-León, Carlos F. and Troup Howard**, “The Assessment Gap: Racial Inequalities in Property Taxation,” *The Quarterly Journal of Economics*, February 2022, *137* (3), 1383–1434.
- Coste, Jonah**, “Capitalization of Property Tax Incentives: Evidence From Philadelphia,” Working Paper 24-01, Federal Housing Finance Agency January 2024. Accepted for publication in *Real Estate Economics*.
- Ferman, Bruno, Cristine Pinto, and Vitor Possebom**, “Cherry Picking with Synthetic Controls,” *Journal of Policy Analysis and Management*, 2020, *39* (2), 510–532.
- McMillen, Daniel and Ruchi Singh**, “Assessment Regressivity and Property Taxation,” *The Journal of Real Estate Finance and Economics*, 2020, *60* (1), 155–169.